

RoomShare Platform - Architecture & Structure Blueprint

1. Executive Platform Summary

RoomShare is a modern, high-performance web platform for residential room and house rentals in Ghana.

It supports dual-role access control for Landlords and Tenants.

Key capabilities: listing management, verified room discovery, direct messaging, end-to-end lease workflow.

2. Technology Stack

- Frontend: React 19 + TypeScript + Vite 6
- Styling: Tailwind CSS v4
- Iconography & Motion: Lucide React icons + Framer Motion
- Backend: Express.js on Node.js (Port 3000)
- Persistence: Server-side JSON DB (data/db.json) + local client sync
- Image Optimizer: Canvas-based client downscaling to ~1400px

3. System Components & Routes

- Room Discovery & Filter Engine (App.tsx & FilterBar.tsx)
- Room Details Modal (RoomDetailModal.tsx)
- Landlord Dashboard & Listing Wizard (LandlordDashboard.tsx)
- Tenant Hub & Tracker (TenantApplicationsView.tsx)
- Real-Time Inbox (MessagesView.tsx)
- Lease Application Modal (ApplicationModal.tsx)

4. REST API Endpoints

- GET /api/data
- POST /api/sync
- GET / POST / PUT / DELETE /api/rooms
- GET / POST / PUT /api/users
- GET / POST / PUT /api/applications
- GET / POST /api/conversations
- GET / POST /api/messages
- GET /website-structure.pdf